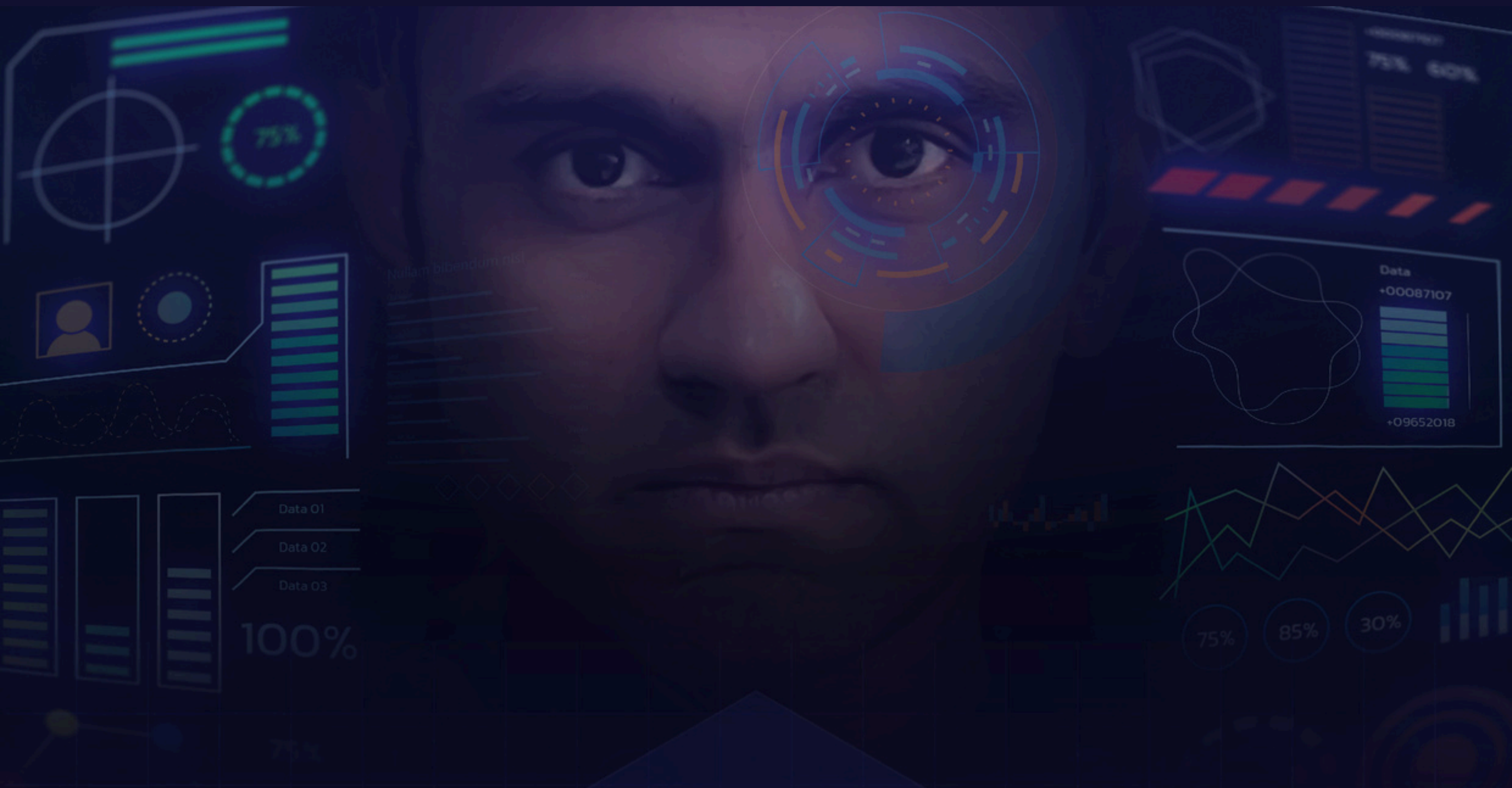

CODE BASICS

Large Language Models

TAKEAWAYS

codebasics.io

How LLMs Work?

1. LLMs are autoregressive models where they predict one token at a time and uses tokens produced from first iteration as an input in the next one to produce new tokens
2. Transformer architecture is the core of any LLM
3. Training an LLM requires huge amount of resources and the cost associated with training is very high

Context Window, Temperature, Top-p and Top-k

1. Context window determines the number of past tokens an LLM considers when generating the next token, influencing coherence and memory.
2. Temperature controls randomness in text generation – higher values produce more creative outputs, while lower values lead to deterministic results.
3. Top-k sampling limits token selection to the top k most probable options, reducing randomness while maintaining diversity.

Context Window, Temperature, Top-p and Top-k

4. Top-p (nucleus) sampling selects tokens from the smallest set whose cumulative probability exceeds p, dynamically adjusting the token pool size.

5. Combining temperature, top-p, and top-k allows fine-tuning of model output, balancing creativity and relevance in text generation.

Gen AI Application Development Steps

1 Gen AI app development steps,

1. Evaluate if it is really a Gen AI app
2. Data collection and preparation
3. Choose right model architecture
4. Model training
5. Evaluation
6. Optimize and deploy
7. Compliance and ethical considerations
8. Monitor and feedback

What is a Vector Database?

1. Vector databases store and manage high-dimensional vector representations of data, enabling efficient similarity search and retrieval.
2. Databases such as Milvus, Pinecone, Qdrant, Chromadb are gaining attention in the modern day Gen AI boom
3. Vector databases excel at handling unstructured data (text, images, audio) by converting it into embeddings using neural networks.
4. They use a combination of different algorithms that all participate in Approximate Nearest Neighbor (ANN) search

What is RAG (Retrieval Augmented Generation)?

1. Retrieval-Augmented Generation (RAG) combines information retrieval with text generation, enhancing LLMs with external knowledge.

2. RAG retrieves relevant documents or data chunks and uses them to generate more accurate and context-aware responses.

3. This approach reduces hallucination by grounding outputs in factual information from the retrieved content.

4. RAG is widely used in question answering, chatbots, and document summarization to improve precision and relevance.

5. By dynamically incorporating up-to-date information, RAG allows LLMs to handle knowledge-intensive tasks beyond their training data.

Gen AI Tech Stack

1. Tools used Gen AI application development
   - Framework: Langchain, HuggingFace, Pytorch etc.
   - Vector databases: Qdrant, ChromaDB, Pinecone etc
   - Cloud: AWS Bedrock, Azure OpenAI service, Google AI studio
   - LLMs: GPT, Gemini, Claude, LlaMa, Mistral etc.
2. Langchain is a framework that makes building Gen AI applications easier and faster